Rising Water, Buried Contamination

Sea-level rise and known contaminated sites in Camden, New Jersey

David Chang · Magnet High School, Class of 2027 · October 2026 · Independent student research, data and code: github.com/davidjchang27/nj-toxic-tides

Key findings

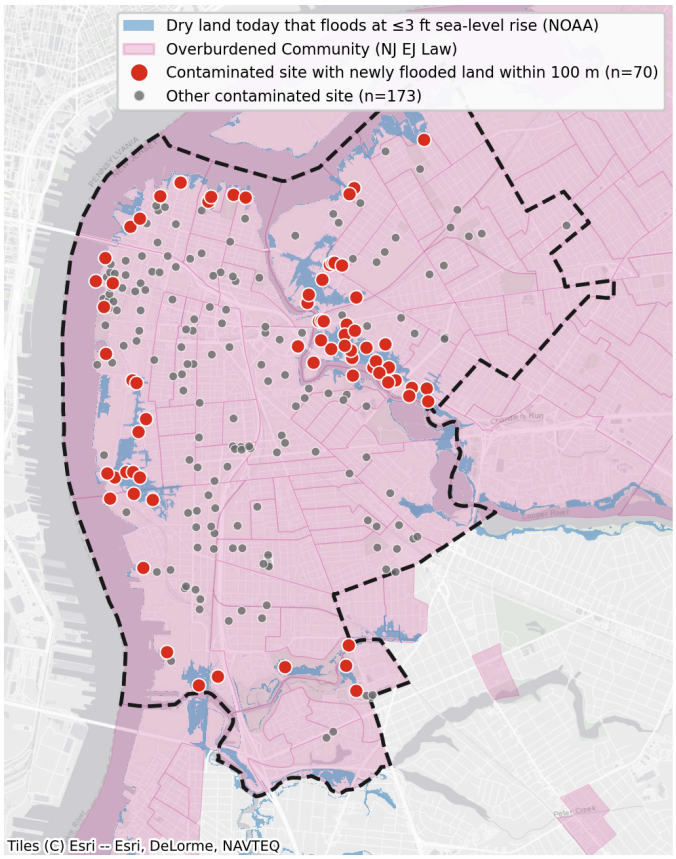
- **70 of Camden's 243 known contaminated sites** have land within 100 m that is dry today but would flood with 3 ft of sea-level rise. For **14** of them, that flooding begins at just **1 ft**, within New Jersey's likely range for 2050 (0.9–1.7 ft). 3 ft is within the likely range for 2100 under intermediate emissions (2.2–3.8 ft).
- **48** of these sites are active cleanups, and **20** have contaminated groundwater under an ongoing Classification Exception Area. 56 of 70 are already in FEMA's 1% annual-chance floodplain.
- About **50,536 residents (70% of the city)** live within 1 km of one of these sites. Every Camden neighborhood is an Overburdened Community under the NJ Environmental Justice Law.
- The largest cluster is the **Admiral Wilson Boulevard / Gateway Park corridor** along the Cooper River: 20 flood-exposed sites, 15 of them active cleanups.

70
contaminated sites with new flooding nearby at ≤3 ft

14
where it starts at 1 ft (2050-range)

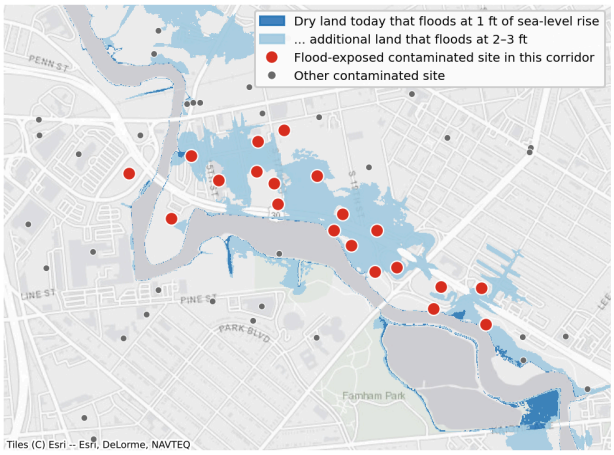
36,815
residents within 1 km of a 1-ft site (51% of Camden)

3.1×
statewide: OBC residents' likelihood of living near such sites vs. other NJ residents



Red: contaminated sites with land within 100 m that floods at ≤3 ft. Blue: land that is dry today but floods at ≤3 ft (NOAA). Pink: Overburdened Community block groups. Site points are approximate address locations.

HOTSPOT: ADMIRAL WILSON BOULEVARD / GATEWAY PARK



Dark blue floods at 1 ft of sea-level rise. Light blue floods at 2–3 ft. This corridor along the Cooper River holds a concentration of active cleanups and redevelopment parcels.

Site (NJDEP KCSL name)	Address	Flooding begins at	Land within 100 m flooded at 3 ft	Cleanup status	Groundwater CEA
Air Products & Chemicals Inc	2710 Broadway	1 ft	93%	Post-remediation	—
Gateway Park Properties Parcel S17	1950 Admiral Wilson Blvd	1 ft	92%	Active	—
Gateway Park Properties Parcel S16	1820 Admiral Wilson Blvd	1 ft	80%	Active	—
Former ABC Companies	1421 1521 Admiral Wilson Blvd	1 ft	71%	Post-remediation	Yes
Gateway Park Properties Parcels S18 & S19	Rear 2004 Admiral Wilson Blvd	1 ft	64%	Active	—
Gateway Park	Admiral Wilson Blvd	1 ft	59%	Post-remediation	—
Camden City Reverend Evers Park	Morgan St & Olive St	1 ft	49%	Active	—

The seven Camden sites where new flooding begins at the lowest sea level. The full list of 70 sites is in the project repository.

WHY IT MATTERS

Many cleanups leave contamination in place under engineered caps, or rely on restrictions on using contaminated groundwater. Such remedies may not have been designed for higher sea levels. Repeated tidal or storm flooding can erode caps, raise groundwater, and carry contaminated soil and water into nearby streets, homes, and the Cooper and Delaware rivers. New Jersey's REAL flood rules, adopted in January 2026, raise building standards for *new* development and substantial improvements in flood areas. Existing contaminated sites raise a separate question: whether their remedies will hold as sea levels rise.

QUESTIONS AND RECOMMENDATIONS

1. **Remedy resilience review.** For the 14 sites where flooding begins at 1 ft, ask NJDEP and the responsible Licensed Site Remediation Professionals whether the remedies account for projected sea-level rise.
2. **Plan integration.** Add these sites to Camden City and Camden County hazard-mitigation and climate plans, so that flood projects along the Cooper River consider contamination.
3. **Post-flood checks.** After major floods, inspect caps and monitoring wells at exposed sites and publish the results.
4. **Community notice.** Share site-level flood exposure with residents in English and Spanish: 23 of the 70 exposed sites are in block groups that also meet the Limited-English criterion.

METHODS AND LIMITATIONS

Sites: NJDEP Known Contaminated Sites List (pulled Oct. 10, 2026). Flooding: NOAA Office for Coastal Management sea-level-rise inundation layers (feet above Mean Higher High Water). Today's tidal water is excluded, so only land that is dry today counts. Communities: NJDEP Overburdened Community layer and the 2020 Census. A site counts as exposed if any land within 100 m of its mapped point floods at ≤3 ft. This is a **screening analysis**. Site points are approximate addresses, the NOAA layers model permanent inundation only (not storm surge, which reaches farther), and the analysis does not predict that any specific site will release contamination. Statewide comparison: OBC residents are 3.1× (95% CI 2.7–3.5) as likely as other residents to live within 1 km of a flood-exposed contaminated site. Projections: NJ Science and Technical Advisory Panel, *New Jersey's Rising Seas and Changing Coastal Storms* (2025). Contact: david.j.chang27@gmail.com.